



# VANSHU GALHOTRA

Master of Computer Applications

Gender: Male

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## Educational Qualification

| Year | Degree/Examination | Institution/Board                  | CGPA/Percentage |
|------|--------------------|------------------------------------|-----------------|
| 2027 | M.C.A              | NIT, Trichy                        | 09.44*          |
| 2024 | B.C.A              | Panjab University, Chandigarh      | 85.91%          |
| 2021 | Class XII          | DAV Public School, Kotkapura, CBSE | 98.20%          |
| 2019 | Class X            | DAV Public School, Kotkapura, CBSE | 90.60%          |

## Academic Achievements

- **BCA (First Position):** DAV College, Chandigarh (Panjab University)
- **1<sup>st</sup> Prize:** Software Development Competition Inter College (Panjab University)
- **2<sup>nd</sup> Prize:** Debugging Competition Inter College (Panjab University)
- **2<sup>nd</sup> Prize:** Mathematics Quiz Intra College

## Internship Experience

- **Summer Internship at Goldman Sachs:** *May 2026 – July 2026*  
**Built an MCP Server for a GenAI Support Assistant:** Designed an MCP server integrating LLMs with AI Agent orchestration, exposing read-only, pre-defined database queries as tools. Implemented **token-based authentication** and **role-based, tool-level authorization**. Developed a prompt-to-tool evaluation pipeline to validate LLM tool selection accuracy, and deployed the application on **Kubernetes**.
- **Research Internship at DCSA, Panjab University, Chandigarh:** *May 2025 – Jun 2025*  
Worked on the project **“Handwritten Trajectories Identification in Scanned Text”**, focused on recovering stroke order from static handwritten images. The project involved implementing **graph-theoretic and heuristic-based methods** to extract plausible pen trajectories. A neural network was used to recognize the recovered feature vector, achieving a recognition accuracy of **98.90%**. The results were at par with existing literature. The work also led to the preparation of a **research paper** during the internship.

## Other Projects

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- **Real time Fleet Management System:** Oct 2025 - Jan 2026  
Designed and developed a production-grade real-time fleet management system for a private logistics company to manage vehicle records, document tracking, and operational workflows. Built using Next.js for the frontend and NestJS for the backend with a PostgreSQL database. Deployed on AWS with Nginx as a reverse proxy, and implemented automated S3 backups for data reliability. The system is currently live and actively used in daily operations.
- **BackTrack – Reverse Engineering Event Platform (INFOTREK'25):** May 2025 - Jun 2025  
Developed **BackTrack**, a web platform created for the INFOTREK'25 event organised by **ACM**. The website is specifically used for managing and hosting the reverse engineering event. Built using **Next.js** for the frontend and **NestJS** for the backend, the project also incorporates **CI/CD pipelines** for automated deployment and smooth development workflow.
- **Python Chess Engine Library (pychess-engine):** Sep 2024 - Dec 2024  
Developed **pychess-engine**, a Python based chess engine library available on **PyPI**. It is designed to evaluate board positions, calculate the best moves, and support integration with various interfaces. The engine follows object-oriented principles for a modular structure, uses alpha-beta pruning for efficient move evaluation, and applies multi-threading to improve performance.

## Areas of Interest

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- Data Structures and Algorithms
- Design and Analysis of Algorithms
- Machine Learning
- Deep Learning

## Technical Skills and Certifications

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- Languages : C, C++, Python, JavaScript, Java (Basic)
- Web & Backend : React.js, Next.js, Node.js, NestJS, HTML, CSS, Tailwind CSS, MongoDB, PostgreSQL
- Cloud & DevOps : AWS, Docker, Kubernetes, Nginx, CI/CD, Vercel
- AI Tooling : Machine Learning, Deep Learning, MCP (Model Context Protocol)
- Others : Git, GitHub, OOP, Data Structures & Algorithms (DSA)
- Certifications : Crash Course on Python (Coursera offered by Google)

## Positions of Responsibility

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- **Chairperson – ACM Student Chapter NIT Trichy:** Oct 2025 - Present  
Responsible for helping organize technical events and workshops under the student chapter. Contributed to the planning of the INFOTREK and other activities aimed at building interest in tech among students.
- **EEC Head, VERSION'26** Nov 2025 - Present  
Managed team to plan, develop, and execute events for VERSION'26.

## Extracurricular Activities

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### Sport Activities:

- Basketball – **Sportsfete'24**, the inter department sports fest organized by NIT Trichy. Supported the department team during the event.